

Question 1 [12 Marks]

Four feet of wire is to be used to form a square (having side 'x') and a circle (of radius 'r'). Determine a function that expresses the combined area of both figures, then express the area in terms of 'x'. What is the domain of this function?

$$\text{Area} = (\text{Area of square}) + (\text{Area of Circle})$$

$$A = x^2 + \pi r^2 \rightarrow \textcircled{a}$$

As total length of wire is 4ft, so

$$4 = (\text{Perimeter of square}) + (\text{Circumference of Circle})$$

$$4 = 4x + 2\pi r \rightarrow \textcircled{b}$$

$$\text{So } r = \frac{2(1-x)}{\pi}$$

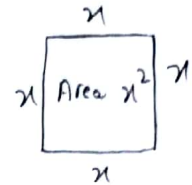
Substituting value of 'r' in $\textcircled{a}$ we get

$$A = x^2 + \pi \left[\frac{2(1-x)}{\pi} \right]^2$$

$$A = x^2 + \frac{4(1-x)^2}{\pi}$$

$$A = \frac{1}{\pi} [(\pi+4)x^2 - 8x + 4]$$

Feasible domain is $0 \leq x \leq 1$ restricted by square's perimeter.



Perimeter : $4x$



Circumference $2\pi r$

Question 2 [13 Marks]

A toy company wants to design a track for a toy car that starts out along a parabolic curve and then converts to a straight line. The function that describes the track is to have the form:

$$f(x) = \begin{cases} \frac{1}{5}x^2 + bx + c & \text{if } x < 4 \\ \frac{1}{3}x + 2 & \text{if } x \geq 4 \end{cases}$$

where x and $f(x)$ are in inches. For the car to move smoothly along the track, the function $f(x)$ must be both continuous and differentiable at 4. Find values of b and c that make $f(x)$ both continuous and differentiable. (Note: Find derivative by definition).

For a fn to be Continuous at $x=4$

$$LHL = RHL = f(4)$$

$$\underline{\underline{LHL}} \quad \lim_{x \rightarrow 4^-} \left[\frac{1}{5}x^2 + bx + c \right] \Rightarrow \frac{16}{5} + 4b + c$$

$$\underline{\underline{RHL}} \quad \lim_{x \rightarrow 4^+} \left[\frac{1}{3}x + 2 \right] \Rightarrow \frac{4}{3} + 2 \Rightarrow \frac{10}{3}$$

$$f(4) = \frac{1}{3}(4) + 2 \Rightarrow \frac{10}{3}$$

So for Continuity

$$\frac{16}{5} + 4b + c = \frac{10}{3}$$

$$4b + c = \frac{10}{3} - \frac{16}{5}$$

$$4b + c = \frac{2}{15}$$

$$\boxed{c = \frac{2}{15} - 4b}$$

For differentiability at $x=4$

$$LHD = RHD$$

$$f'(4) = \lim_{z \rightarrow 4^-} \left[\frac{\frac{1}{5}z^2 + bz + c}{z-4} \right] - \frac{10}{3}$$

$$= \lim_{z \rightarrow 4^-} \frac{\frac{1}{5}z^2 + bz + \frac{2}{15} - 4b - \frac{10}{3}}{z-4}$$

$$= \lim_{z \rightarrow 4^-} \frac{\frac{1}{5}z^2 + bz - 4b - \frac{16}{5}}{z-4}$$

$$= \lim_{z \rightarrow 4^-} \frac{z^2 + 5bz - 20b - 16}{5(z-4)}$$

$$= \lim_{z \rightarrow 4^-} \frac{(z-4)(z+4+5b)}{5(z-4)} \Rightarrow \frac{4+4+5b}{5} \Rightarrow \frac{8+5b}{5} \text{ or } \frac{8}{5} + b$$

$$\text{RHD } \lim_{z \rightarrow 4^+} \frac{\frac{1}{3}z + 2 - \frac{10}{3}}{z-4}$$

$$\lim_{z \rightarrow 4^+} \frac{\frac{1}{3}z - \frac{4}{3}}{z-4}$$

$$\lim_{z \rightarrow 4^+} \frac{z-4}{3(z-4)} \Rightarrow \frac{1}{3}$$

As LHD = RHD

$$\frac{8}{5} + b = \frac{1}{3}$$

$$b = \frac{1}{3} - \frac{8}{5} = -\frac{19}{15}$$

$$c = \frac{2}{15} + 4\left(\frac{-19}{15}\right)$$

$$c = \frac{26}{5}$$

So value of $b = -\frac{19}{15}$ & $c = \frac{26}{5}$
in order to make $f(x)$ continuous & diff at $x=4$.